A NOETHERIAN RING IS A G-RING EXACTLY WHEN EVERY FINITE FREE EXTENSION HAS REGULAR FORMAL FIBRES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- G-ring condition tested through formal fibres and purely inseparable residue field extensions. the discussion before the statement, source lines 12179-12261. Supplies the criterion the proof verifies.
- Existence of a finite free extension realising a prescribed finite residue field extension at a prime lying over a given prime. the author cites Algebra, Lemma algebra-lemma-finite-free-given-residue-field-extension at proof lines 12275-12278. Produces the extension used in the computation.
- Completion of a finite ring extension decomposes as a product of the completions at the primes lying over the given prime. the author cites Algebra, Lemma algebra-lemma-completion-finite-extension at proof lines 12279-12283. Gives the displayed product decomposition from which regularity is read off.

Source reference. Original lemma 07PQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9805–9805. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07NE: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9759–9759. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07N9: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9754–9754. Navigation pointer to a statement the

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Source reference. Original lemma 07PP: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9804–9804. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a Noetherian ring. Then R is a G-ring if and only if for every finite free ring map $R \to S$ the formal fibres of S are regular rings.*

Proof. Assume that for any finite free ring map $R \to S$ the ring S has regular formal fibres. Let $\mathfrak{q} \subset \mathfrak{p} \subset R$ be primes and let $\kappa(\mathfrak{q}) \subset L$ be a finite purely inseparable extension. To show that R is a G-ring it suffices to show that

$$R^\wedge_\mathfrak{p} \otimes_R \kappa(\mathfrak{q}) \otimes_{\kappa(\mathfrak{q})} L$$

is a regular ring. Choose a finite free extension $R \to R'$ such that $\mathfrak{q}' = \mathfrak{q}R'$ is a prime and such that $\kappa(\mathfrak{q}')$ is isomorphic to L over $\kappa(\mathfrak{q})$, see Algebra, Lemma 07NE. By Algebra, Lemma 07N9 we have

$$R^\wedge_\mathfrak{p} \otimes_R R' = \prod (R'_{\mathfrak{p}'_i})^\wedge$$

where $\mathfrak{p}'_i$ are the primes of R' lying over $\mathfrak{p}$. Thus we have

$$R^\wedge_\mathfrak{p} \otimes_R \kappa(\mathfrak{q}) \otimes_{\kappa(\mathfrak{q})} L = R^\wedge_\mathfrak{p} \otimes_R R' \otimes_{R'} \kappa(\mathfrak{q}') = \prod (R'_{\mathfrak{p}'_i})^\wedge \otimes_{R'_{\mathfrak{p}'_i}} \kappa(\mathfrak{q}')$$

Our assumption is that the rings on the right are regular, hence the ring on the left is regular too. Thus R is a G-ring. The converse follows from Lemma 07PP. $\square$